Sep 29, 2026 (Due: 08:00 Oct 13, 2026)

1. Let

$$A = \begin{bmatrix} \alpha_1 & \eta_2 & \eta_3 & \cdots & \cdots & \eta_n \\ \xi_2 & \alpha_2 & 0 & \cdots & \cdots & 0 \\ \xi_3 & 0 & \alpha_3 & \ddots & & \vdots \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \vdots & & \ddots & \alpha_{n-1} & 0 \\ \xi_n & 0 & \cdots & \cdots & 0 & \alpha_n \end{bmatrix} \in \mathbb{R}^{n\times n}.$$

Design an efficient algorithm to compute the QR factorization of A.

2. Let $A \in \mathbb{C}^{m\times n}$, $b \in \mathbb{C}^m$. Show that solving the least squares problem

$$\min_{x\in\mathbb{C}^n} \|b - Ax\|_2$$

is mathematically equivalent to solving the *augmented* system

$$\begin{bmatrix} I_m & A \\ A^* & 0 \end{bmatrix} \begin{bmatrix} r \\ x \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix}.$$

3. Implement the algorithm we discussed in the lecture for solving the least squares problem with equality constraints (LSE), and test it on some randomly generated problems. Make sure your algorithm does *not* explicitly form the orthogonal/unitary factor in the elimination stage.

(optional) Can you construct an equivalent augmented form for this problem, that is analogous to Exercise 2?

4. Write a program to compute the QR factorization of a complex matrix $A \in \mathbb{C}^{n\times n}$ with
(1) Cholesky-QR;
(2) Householder-QR;
(3) CGS;
(4) MGS;
(5) CGS2;
(6) MGS2.
Visualize the *componentwise* loss of orthogonality $|Q^*Q - I_n|$ and the residual norm $\|A - QR\|_{\mathsf{F}}$ with a few well-conditioned and ill-conditioned examples. If you use MATLAB/Octave, you may find `imagesc()` helpful.

5. Find the "best" straight line that approximately passes through the data set $\{(n, \ln n) \in \mathbb{R}^2 \colon n \in \{3, 4, \ldots, 10\}\}$. Visualize your result and clarify in what sense your solution is the best.

6. (H) Use pseudocode to describe a block algorithm (either left-looking or right-looking) for Householder-QR. Make sure all indices are correct. For simplicity, you may assume that the number of columns of the matrix is a multiple of the block size.

7. (optional) Given α, $\beta \in \mathbb{C}$. Design an algorithm to compute $c \in \mathbb{R}$ and $s \in \mathbb{C}$ such that

$$\begin{bmatrix} c & -\overline{s} \\ s & c \end{bmatrix}^* \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} * \\ 0 \end{bmatrix} \quad \text{and} \quad c^2 + |s|^2 = 1.$$

8. (optional) Let w_1, w_2, …, $w_k \in \mathbb{C}^n$ be unit vectors. Try to find a matrix $T \in \mathbb{C}^{k \times k}$ such that

$$(I - 2w_1 w_1^*)(I - 2w_2 w_2^*) \cdots (I - 2w_k w_k^*) = I - [w_1, w_2, \ldots, w_k] T [w_1, w_2, \ldots, w_k]^*.$$

9. (H, optional) Implement the Bunch–Kaufman algorithm to solve indefinite linear systems. Test your implementation using a few examples with known solutions. Are the backward errors consistent with the theoretical prediction?